Sarcasm Detection using NLP Approach

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1 Overview

My final project focuses on using NLP approach to identify sarcasm. The labeled dataset I use is News Headlines dataset for Sarcasm Detection. This dataset is a compilation of 26709 news headlines. Approximately 56.1% of these headlines are nonsarcastic headlines, with the remaining 43.9% being sarcastic. As the dataset is slightly imbalanced, I address the issue of class imbalance in the later fine-tuning process. The initial data is split into training, validation and test sets with shuffling at the ratio of $0.7 : 0.15 : 0.15$.

2 Approaches

Three approaches are employed in this final project: feature extraction, fine-tuning (including both full fine-tuning and fine-tuning a limited set of parameters), and utilizing fine-tuned models available from Hugging Face. The pretrained model selected for both feature extraction and fine-tuning is MiniLM-L12-H384-uncased.

2.1 Feature Extraction

For feature extraction, three different post-processing methods are explored: averaging over token dimensions, retaining the hidden state vector for the initial token, and preserving the hidden state vector for the last valid token. Within each post-processing approach, I tune the hyperparameters for Logistic Regression and Random Forest models separately. This leads to a total of six unique combinations (3 post-processing approaches $\times$ 2 classification models $= 6$).

Hyperparameter tuning reveals that optimal hyperparameters for the same classification model may vary across different post-processing approaches. Consequently, all six models, each with its corresponding tuned hyperparameters, are applied to the test dataset, and their performance is detailed in Section 3.

2.2 Fine-tuning

2.2.1 Full Fine-tuning

To address slight data imbalance in class, class weights are incorporated into the loss function during model fine-tuning.

2.2.2 Fine-tuning with PEFT and LoRA

To explore the potential of achieving performance similar to full fine-tuning with a minimal number of trainable parameters, PEFT approaches are implemented using the PEFT library. These approaches fine-tune only a small fraction of model parameters while keeping the majority of pretrained model parameters frozen. In my project, approximately 44% of all parameters are made trainable. As demonstrated in Section 3, this approach achieves performance on par with full fine-tuning.

2.3 Using Fine-tuned Model from Hugging Face

The fine-tuned model from Hugging Face I use is jkhan447/sarcasm-detection-RoBerta-base-CR.

3 Performance Comparison

Table 1 presents accuracy, precision (macro average), recall (macro average), and F1-score (macro average) for various models. Of all the approaches, full fine-tuning stands out with the highest performance, achieving an F1-score of 0.92. This is not surprising as full fine-tuning allows the model to adapt and specialize to the specific task of sarcasm detection by fine-tuning all of its parameters. This extensive fine-tuning process enables the model to capture intricate patterns and nuances in the data, resulting in superior performance. Fine-tuning with PEFT and LoRA achieves a similarly strong performance. In the category of models created through feature extraction, the combination of averaging over token dimensions and Logistic Regression attains the top F1-score of 0.88.

Table 1: Performance evaluation of different approaches

Approach	Accuracy	Precision	Recall	F1-Score
Feature extraction (average over token dimension + Random Forest)				0.82
Feature extraction (keep last + Random Forest)				0.81
Feature extraction (keep first + Random Forest)				0.80
Feature extraction (average over token dimension + Logistic)				0.88
Feature extraction (keep last + Logistic)				0.85
Feature extraction (keep first + Logistic)				0.85
Full fine-tuning	0.93	0.93	0.92	0.92
Fine-tuning with PEFT and LoRA	0.92	0.92	0.91	0.92
Fine-tuned model from Hugging Face	0.50	0.50	0.50	0.50